Weather, Interconnections, and Energy Prices

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Abstract

We construct an auditable European electricity-price dataset and reproduce a teammate's XGBoost pipeline. Controlled France experiments identify prior-day auction prices as the largest measured feature improvement; post-origin realised weather adds little. These conditional benchmarks establish a stronger baseline for subsequent graph experiments; historical publication vintages remain unverified.

1 Approach

At 12:00 Europe/Brussels on calendar day $D-1$, predict every hourly day-ahead price of delivery day D, including 23/25-hour daylight-saving days. Our planned graph connects bidding zones, interconnectors and JAO transmission constraints. We implement a dataset builder and per-zone XGBoost [1], and reproduce the teammate's `c1_xgb_zone` pipeline unchanged at `757da793`.

2 Experiments

Exact data. Our master has 5,887,954 zone-hours, 62 price-area identities, January 2016–October 2026. ENTSO-E prices, load forecasts and physical flows use releases v1.2. We join PECD/ERA5 temperature, anomaly and wind/solar capacity factors (2010–September 2026) and original CWE/CORE JAO exports (May 2015–October 2026). Weather is reanalysis; its v1.1 release identity is unverified. We retain NaNs, negative prices and separate DE_AT_LU/DE_LU identities. The partner's original v1.0 table has 4,240,035 rows and 45 zones through September 2026, with filled inputs and a historical German splice. The appendix records hashes and geography.

Splits and metrics. Fit on January 1, 2016–March 31, 2024 (pretrain + train); validate on April 1, 2024–March 31, 2025; test on April 1, 2025–September 30, 2026, using Brussels delivery dates. October 2026 remains unassigned. MAE and RMSE are in EUR/MWh. Weekly persistence copies the same zone's price 168 elapsed hours earlier. Comparisons use identical valid target-hour masks, never feature-complete subsets or filled labels. The ten zones BE, FR, NL, DE_LU, AT, ES, PT, PL, CZ, CH supply 87,600 validation and 131,520 test zone-hours per method.

Models and aggregate results. Our inputs are hour, weekday, month, UTC offset, CWE/CORE era flags and weekly price. Switching raw-price squared error to $z = \operatorname{asinh}(p/50)$ with L1 loss improves test MAE; invert with $50\sinh(\hat{z})$ before scoring. The partner adds prior-day price, load forecast, four weather lags and net imports, and changes tree settings (Table 1). Its full 43-zone reproduction scores 21.25 test MAE versus the published 21.24.

Method (same ten zones)	Validation MAE	Test MAE
Weekly persistence	31.24	29.15
Calendar + weekly, raw-price MSE	29.65	29.59
Same inputs, asinh + L1	29.44	28.81
Reproduced partner, richer inputs	21.04	21.17

Table 1: Matched ten-zone MAE (EUR/MWh). All models fit per zone.

Controlled France experiments. Fix the partner learner: CPU histogram trees, depth 6, at most 600 rounds, learning rate 0.05, row/column sampling 0.8, L1 objective, asinh scale 50, seed 0, and 50-round early stopping on transformed validation MAE. Retain all 72,310 training, 8,760 validation and 13,152 test targets.

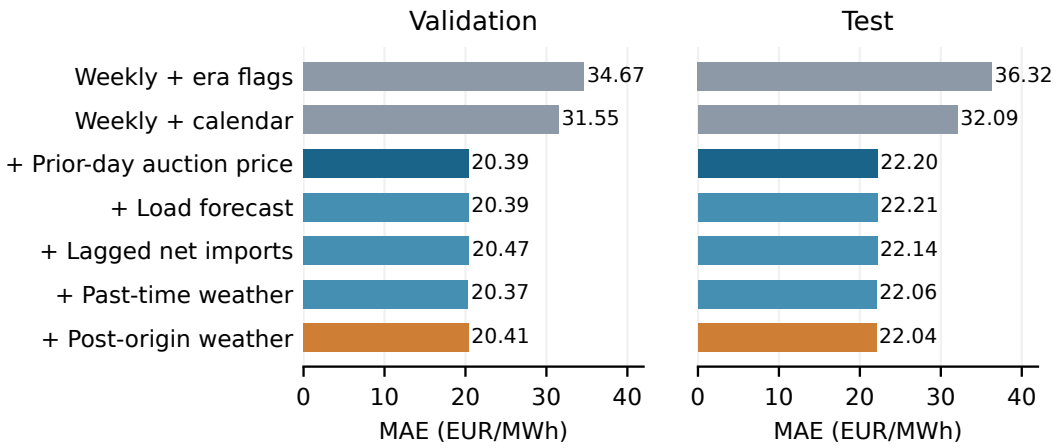


Figure 1: France, fixed learner and targets. Past-time weather masks lag-24 intervals ending after the origin; the last row unmasks them. Later rows guard prior-day prices by delivery date.

Changing only our seven-feature model to partner hyperparameters worsens test MAE to 36.32. Its equivalent legacy-table control matches within 10^{-6} EUR/MWh. Removing the era/offset inputs and adopting the partner feature order gives 32.09. Adding prior-day auction prices then reduces test MAE to 22.20 (30.8%); the validation improvement is 11.16 EUR/MWh (95% paired calendar-week bootstrap interval [8.55, 14.13], 2,000 replicates). This is the largest measured addition; other increments are small and mixed.

Timing audit and interpretation. Previous-day day-ahead prices are normally auctioned on $D-2$ [2]; their delivery time does not imply unavailability. However, an elapsed 24-hour lag can reach day D on the 25-hour autumn transition; we mask that case (one validation and one test hour). Half the original weather-lag-24 observations occur after the origin. Masking them gives 22.06 test MAE, versus 22.04 when supplied; using 48-hour weather lags gives 22.12. Post-origin weather therefore does not explain most of the France advantage. Past effective time still cannot prove publication availability: ERA5 is retrospective [3], and load/weather revision vintages are absent. These are conditional scores, not certified operational forecasts.

Limitations. This one-country, one-seed sequence is order-dependent, not causal feature importance. Bootstrap intervals condition on fitted models. Repeated exploratory test inspection prevents a fresh blind-confirmation claim. Detailed CNEC and graph models remain future work.

3 Future work

Establish a publication-aware recent-price baseline, obtain archived weather forecasts, and repeat across countries and seeds. Compare neighbour features, interconnector graphs and JAO constraint nodes under identical evaluation rules, keeping physical flows, scheduled exchanges and MaxBex distinct.

References

[1] T. Chen and C. Guestrin. XGBoost: A scalable tree boosting system. *KDD*, 2016. arXiv:1603.02754.

[2] Nord Pool. Day-ahead market: auction timeline. Accessed November 2026.

[3] ECMWF. ERA5 global atmospheric reanalysis. Accessed November 2026.